

This is the description of the work plan for WG II.

The goal is to generate initial conditions for an isolated MW-like galactic disk. We use for that purpose an older version of the MakeDisk code written by Volker Springel to generate quasi-equilibrium 4 component systems (dark halo, stellar bulge, stellar disk, gas disk). We will provide on a web repository several files containing initial particle coordinates and velocities that everybody will be able to download. Here is a short description of the data we will provide.

UNITS

Velocity: km/s
Mass: 10^9 Msun
Length: kpc

Disk properties (in above units)

Disk scale length: 3.43218
Disk scale height: 0.343218 (10% of scale length)
M_DISK: 42.9661
M_GAS: 8.59322
M_200: 1074.15
R_200: 205.479
Total mass: 1301.7

For other quantities, see the file main.c in the MakeDisk package.

Number of particles and particle mass resolution

We will work at 3 different mass resolution, namely $1e5$, $1e6$ and $1e7$ particles per main component. For the lowest resolution case, we have for example:

NGAS= $1e5$ ----> $m_{gas_res} = 8.593e4$ Msun
NHALO= $1e5$ ----> $m_{dark_res} = 1.254e7$ Msun
NDISK= $1e5$ ----> $m_{disk_res} = 3.4373e5$ Msun
NBULGE=1250 ----> $m_{bulge_res} = 3.4373e5$ Msun (Bulge to total disk ratio=0.1, Bulge to stellar disk=0.125)

Structure of ASCII files

Gas particle (gas.dat): x,y,z,vx,vy,vz, m_{gas} , u_{gas}
Dark matter halo (halo.dat): x,y,z,vx,vy,vz, m_{dark}
Stellar disk (disk.dat): x,y,z,vx,vy,vz, m_{disk}
Stellar bulge (bulge.dat): x,y,z,vx,vy,vz, m_{bulge}

Note that u_{gas} is the specific internal energy of the gas, see Springel (2000). The initial specific energy has been computed by MakeDisk to ensure a global equilibrium. We suggest here that all codes use 10^4 K as initial temperature instead.

For AMR codes

The gas can be initialized analytically following the profile adopted by MakeDisk:

$$\rho(r,z) = \rho_0 \exp(-r/r_d) \exp(-|z|/z_d)$$

where

$$\rho_0 = M_{GAS} / (4 \cdot \pi \cdot r_d^2 / z_d)$$

The scale radius r_d , scale height z_d and mass M_{GAS} of the gas disk are provided in the "Disk properties" section. In RAMSES we truncate the disk at a maximum radius r_{max} and z_{max} , where we typically choose $r_{max}=20$ kpc and $z_{max}=3$ kpc. If chosen large enough, the results are very insensitive to the latter parameters.

In RAMSES we impose a very tenuous gas halo of uniform density ($d_{\text{halo}} \sim 10^{-6} \text{ cm}^{-3}$) such that $M_{\text{halo}} = d_{\text{halo}} * L_{\text{box}}^3 < 1\% M_{\text{DISK}}$. The initial temperature can be set to 10^4 K too, and the gas velocity should be zero in the halo.

The circular velocity profile of the system is provided in the file "vcirc.dat" where the first column is the radius in kpc and the second vcirc in km/s. This circular velocity has to be used to set your disk initial rotational velocity profile in centrifugal equilibrium.

Suggested run parameters:

For all codes, the gas mass resolution has to be as close as possible to the one quoted above. Using a Jeans mass 64x larger and the corresponding Jeans length, this results in a well defined spatial resolution (from 80 pc down to 5 pc). We suggest that all codes, if possible, use this spatial resolution, or as close to it as possible. We also encourage, when applicable, the use of a polytropic pressure floor, so that the Jeans length is properly resolved at all time. The polytrope is defined as $T=T_J (n_H/n_J)$ where T_J and n_J are listed in the table below, together with the corresponding mass and spatial resolution. We also suggest to use cooling at solar metallicity, with both atomic and fine structure (low T) cooling processes, if possible.

Resolution:

$N_{\text{gas}}=1d5$	$m_{\text{gas_res}}=8.593d4 \text{ msun}$	$\Delta x=80 \text{ pc}$	$T_J=1800 \text{ K}$	$n_J=8 \text{ H/cc}$
$N_{\text{gas}}=1d6$	$m_{\text{gas_res}}=8.593d3 \text{ msun}$	$\Delta x=20 \text{ pc}$	$T_J=700 \text{ K}$	$n_J=50 \text{ H/cc}$
$N_{\text{gas}}=1d7$	$m_{\text{gas_res}}=8.593d2 \text{ msun}$	$\Delta x=5 \text{ pc}$	$T_J=300 \text{ K}$	$n_J=300 \text{ H/cc}$

The repository address will be given as soon as possible.